BCD and Gray code

BCD (Binary coded Decimal)

The 8421 code is a type of BCD (binary coded decimal) code. Binary coded decimal means that each decimal digit, 0 through 9, is represented by a binary code of four bits. The designation 8421 indicates the binary weights of the four bits (2^3, 2^2, 2^1, 2^0). The ease of conversion between 8421 code numbers and the familiar decimal numbers is the main advantage of this code.

When we refer to BCD, we always mean the 8421 code unless otherwise stated.

The goal of BCD is to make it simpler to convert human numbers into values that a machine can understand.

It's used in **Embedded systems**.

Invalid codes:

You should realize that, with four bits, sixteen numbers (0000 through 1111) can be represented but that, in the 8421 code, only ten of these are used. The six code combinations that are not used—1010, 1011, 1100, 1101, 1110, and 1111—are invalid in the 8421 BCD code.

Example 1:

Convert each of the following decimal numbers to BCD:

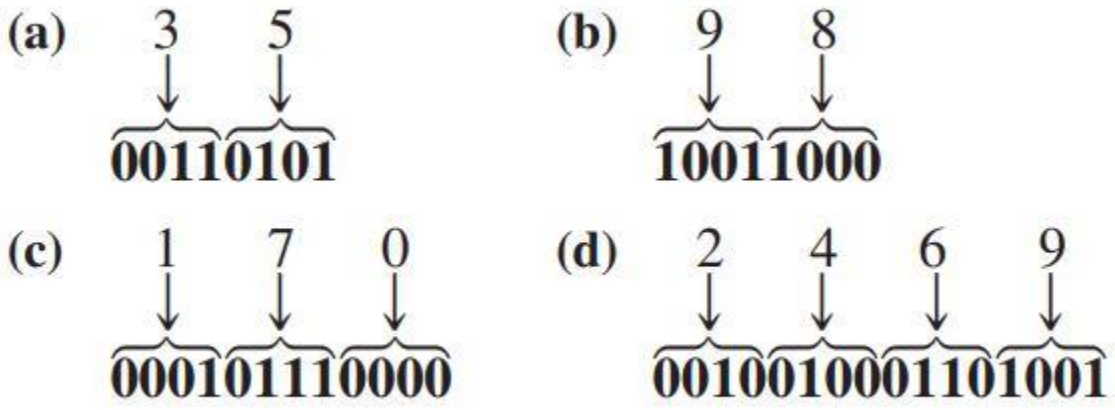
(a) 35

(b) 98

(c) 170

(d) 2469

Answers:



Example 2:

Convert each of the following BCD codes to decimal:

(a) 10000110

(b) 001101010001

(c) 1001010001110000

Answers:

